ChemistrySelect

Research Article
doi.org/10.1002/slct.202304748

Chemistry Europe
European Chemical Societies Publishing

www.chemistryselect.org

A Facile Route Preparation of Fe_3O_4/MWCNT/ZnO/PANI Nanocomposite and its Characterization for Enhanced Microwave Absorption Properties

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With increasing concerns regarding the potential health risks associated with microwave-based technology, researchers have been actively investigating materials with impressive microwave absorption properties. In this study, a synthesis and investigation of a nanocomposite, named Fe_3O_4/MWCNT/ZnO/PANI, consisting of magnetite (Fe_3O_4), multi-walled carbon nanotube (MWCNT), zinc oxide (ZnO), and polyaniline (PANI) as a microwave-absorbing material, was carried out for the first time. The nanocomposite was prepared through a simple physical mixing approach and demonstrated tunable microwave absorption, making it highly suitable for shielding applications. Structural analysis techniques, including X-ray diffraction (XRD), Fourier-transform infrared spectroscopy (FTIR), and scanning electron microscopy with energy-dispersive X-ray analysis (SEM-EDAX), confirmed the presence of all components in the nanocomposite. Moreover, ultraviolet-visible spectroscopy (UV-Vis) further verified the formation of nanocomposite components and revealed an adjustable optical bandgap ranging from 2.86 to 3.34 eV. Impressively, the nanocomposite exhibited promising microwave absorption properties, with a minimum reflection loss (RL) of −32.66 dB for the nanocomposite containing 37.50% PANI by weight, and a maximum electromagnetic bandwidth (EB) of 2.26 GHz for the nanocomposite containing 44.44% PANI by weight, indicating its potential as an effective shielding material. Furthermore, a modified theory combining transmission line, Landau-Lifshits, and Drude-Lorentz theories was introduced to extract complex relative permittivity and permeability, providing insights into the role of polyaniline in enhancing the microwave absorption properties of the nanocomposite as measured by reflection loss (RL).

Introduction

Recently, the use of advanced military equipment has become more prevalent because of ongoing conflicts worldwide. However, there is a disadvantage to this advanced equipment: electromagnetic waves are widely distributed in the microwave spectrum i.e., microwaves, which could harm human health. To address this issue, researchers have explored various shielding mechanisms, including developing materials that can absorb microwaves.[1–3] These materials include graphene-based composites,[4–6] ferrite-based composites,[7–9] and conductive polymer-based composites.[10,11] The objective of this development is to reduce the reflection loss (RL) and enhance the electromagnetic bandwidth (EB) absorption of a material. A desirable microwave absorber should ideally have an RL below −10 dB, which can absorb approximately 90% of electromagnetic waves.[12,13] Therefore, a wider EB with an RL below −10 dB indicates a high-quality microwave absorber.

In contrast to magnetic or semiconductor materials, the performance of absorbing materials cannot be fully explained by only observing a single property. Instead, it is the combination of different properties that make an absorbing material effective. Achieving the right combination of properties is crucial to enhancing the performance of the material. One common approach is to use nanocomposites, which are materials that combine two or more substances at the nanoscale level. Researchers have successfully synthesized the multi-walled carbon nanotube (MWCNT)/zinc oxide (ZnO) nanocomposite using the high-energy milling method, which has demonstrated a minimum RL of −26.4 dB and an EB below −10 dB at frequencies greater than 1.5 GHz.[14] Another study used the high-energy milling method to prepare a nanocomposite containing polyaniline (PANI) as the conductive material and magnetite (Fe_3O_4) as the magnetic material. This resulted in an RL of −28.67 dB, which was deemed satisfactory, with a bandwidth below −10 dB at approximately 1 GHz.[15] Additionally, Fe_3O_4 was used as an additional component in the nanocomposite along with MWCNT to improve the microwave

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absorption properties of the material. Consequently, the Fe_3O_4/MWCNT nanocomposite exhibited an exceptional RL of −34 dB, with a corresponding bandwidth below −10 dB at 0.4 GHz.[16] Moreover, by combining Fe_3O_4 with the ZnO core-shell in a nanocomposite using the hydrothermal method, an impressive RL of −50.79 dB was achieved, with a corresponding bandwidth below −10 dB at approximately 1.5 GHz.[17] The nanocomposite comprising Fe_3O_4, MWCNT, and ZnO achieved a remarkable RL of −34.2 dB, with a corresponding bandwidth below −10 dB at 1 GHz.[18] However, during the course of our investigations, we observed that certain materials required complex and costly preparation procedures, whileothers were easier to prepare but demonstrated inferior microwave absorption properties. Interestingly, polyaniline, a relatively simple material to prepare, demosntrated remarkable enhancement when incorporated as a component in the nanocomposite for microwave absorption applications.[19–21] This finding highlights the potential of polyaniline as a promising material for designing cost-efficient and effective microwave absorbers.

In this study, we propose a simple and effective method for fabricating a microwave absorber material known as the Fe_3O_4/MWCNT/ZnO/PANI nanocomposite, with varying PANI content. This nanocomposite combines magnetic, dielectric, and conductive properties to optimize microwave absorption. We carefully selected Fe_3O_4, ZnO, MWCNT, and PANI as components based on their respective properties, ensuring they work together synergistically within the nanocomposite. Notably, we found that PANI plays a crucial role in tuning the microwave absorption and optical band gap of the nanocomposite. To make the process more cost-effective, we utilized Indonesian natural iron sand as the primary raw material for synthesizing Fe_3O_4, taking advantage of the region's natural resources. Additionally, we developed a modified model that effectively reveal the influence of PANI composition on the nanocomposite. Remarkably, the modified model successfully extracts the reflection loss curve into complex relative permittivity and permeability enabling the investigation of the role of polyaniline in the microwave absorption properties of Fe_3O_4/MWCNT/ZnO/PANI nanocomposite.

Results and Discussion

Formation mechanism

The possible formation mechanism of all components in the Fe_3O_4/MWCNT/ZnO/PANI nanocomposite is represented by the following chemical reactions:

$$\mathrm{MWCNT} + \mathrm{HNO_3} \xrightarrow{\textit{Sonication}} \mathrm{MWCNT\text{-}NO_2} + \mathrm{H_2O}$$

$$8\ \mathrm{NH_4OH} \longrightarrow 8\ \mathrm{NH_4^+} + 8\ \mathrm{OH^-}$$

$$2\ \mathrm{Fe^{3+}} + \mathrm{Fe^{2+}} + 8\ \mathrm{HCl} \xrightarrow{\textit{Stirring}} 2\ \mathrm{FeCl_3} + \mathrm{FeCl_2} + 8\ \mathrm{H^+}$$

$$2\ \mathrm{FeCl_3} + \mathrm{FeCl_2} + 8\ \mathrm{NH_4^+} + 8\ \mathrm{OH^-} + \mathrm{MWCNT\text{-}NO_2} \xrightarrow{\textit{Co-Precipitation}} \mathbf{Fe_3O_4/MWCNT} + 4\ \mathrm{H_2O} + 8\ \mathrm{NH_4Cl}$$

$$\mathrm{Zn(CH_3COO)_2} + 2\ \mathrm{NH_4OH} \xrightarrow{\textit{Co-Precipitation}} \mathrm{Zn(OH)_2} + 2\ \mathrm{NH_4CH_3COO}$$

$$\mathrm{C_6H_5NH_2} + \mathrm{HCl} \longrightarrow \mathrm{C_6H_5NH_3^+} + \mathrm{Cl^-}$$

$$4n\ \mathrm{C_6H_5NH_3^+} + 4n\ \mathrm{Cl^-} + n\ \mathrm{(NH_4)_2SO_8} \xrightarrow{\textit{Stirring}} \mathbf{PANI\text{-}ES} + 2n\ \mathrm{(NH_4)_2SO_4} + n\ \mathrm{H_2}$$

$$\mathbf{PANI\text{-}ES} = \left[-(\mathrm{C_6H_4NH})_2-(\mathrm{C_6H_4N})_2- + 2\mathrm{H^+} + 2\mathrm{Cl^-}\right]_n$$

The initial stage of the nanocomposite preparation involved functionalizing MWCNT using nitric acid through a nitration mechanism, as mentioned in the above chemical reaction. This process resulted in the introduction of nitro ($-NO_2$) groups onto the surface of the MWCNT. The presence of these nitro groups enhance the reactivity of the surface, facilitating stronger bonding withother molecules during the nanocomposite formation.[22] Furthermore, iron chloride was prepared by mixing iron powder with HCl and used as a precursor substance for Fe_3O_4 nanoparticle formation. The Fe_3O_4/MWCNT nanocomposite was obtained by utilizing the stronger bonding ability of functionalized MWCNT. In this process, functionalized MWCNT was sonicated with iron chloride to promote interaction between Fe^{3+} and Fe^{2+} ions and MWCNT-NO_2. The subsequent addition of NH_4OH to the solution resulted in the precipitation of the Fe_3O_4/MWCNT nanocomposite, as explained in the previous equation. However, it's important to note that there is a possibility of an excess of Cl^- ions, as indicated in the formation equation of the Fe_3O_4/MWCNT nanocomposite. This excess of Cl- ions can potentially impact the permittivity of the nanocomposite, which is directly related to its microwave absorption properties. Therefore, a magnetic separation technique was employed in the cleaning step to attract the magnetic material, Fe_3O_4, in such a way that it is separated from the bulk solution containing the excess of Cl^-.

Furthermore, a two-step process was used for the preparation of ZnO nanoparticles. First, NH_4OH was added dropwise to a zinc acetate solution, resulting in the precipitation of zinc hydroxide. The zinc hydroxide was then heated to produce ZnO nanoparticles, as explained in the above equation. In the same separated treatment, PANI-ES was synthesized using ammonium persulfate ($(NH_4)_2SO_8$) as an oxidizing agent. Aniline, a precursor material, was first dissolved in HCl to create a charge carrier in PANI-ES. Then, the sulfur atom underwent oxidation by reacting the modified aniline with ammonium persulfate, forming PANI-ES. This salt comprises an equal number of benzenoid (B, $-(C_6H_4NH)_2-$) and quinonoid (Q, $-(C_6H_4N)_2-$) ring structures, along with excess charge carriers, as described in the above equation. In the final preparation step, a sonochemical method was used to fabricate a Fe_3O_4/MWCNT/ZnO nanocomposite. The dissolved Fe_3O_4/MWCNT nanocomposite underwent sonication to promote interfacial bonding withother materials. Subsequently, ZnO was introduced into the sonicated Fe_3O_4/MWCNT nanocomposite through stirring, facilitating interfacial and electrostatic interactions between ZnO and Fe_3O_4/MWCNT nanocomposite. This process resulted in the formation of the Fe_3O_4/MWCNT/ZnO nanocomposite. Addition-

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ally, the Fe_3O_4/MWCNT/ZnO/PANI nanocomposite was created by mixing the Fe_3O_4/MWCNT/ZnO nanocomposite with different amounts of PANI-ES (16.67%, 28.57%, 37.50%, 44.44%) through stirring. This process resulted in a visible color change to dark blue. The interaction mechanism between Fe_3O_4/MWCNT/ZnO and PANI involves electrostatic interactions, hydrogen bonding, and van der Waals forces, allowing for the adsorption or coating of polyaniline onto the Fe_3O_4/MWCNT/ZnO nanocomposite. The resulting nanocomposites were named FMZP1, FMZP2, FMZP3, FMZP4, corresponding to the PANI-ES composition in the nanocomposite (16.67%, 28.57%, 37.50%, 44.44%).

Structural and morphological analysis of Fe_3O_4/MWCNT/ZnO/PANI nanocomposite

Figure 1(a) displays the XRD pattern of all samples. Each material displays prominent diffraction peaks. Specifically, Fe_3O_4/MWCNT has notable peaks at 2θ values of 26.21°, 30.05°, 32.61°, 35.52°, 43.22°, 53.78°, 57.26°, and 62.73°. These peaks are consistent with the ICSD-36314 corresponding to the cubic phase of Fe_3O_4 and the XRD pattern of MWCNT from the previous result.[23] The presence of clear peaks corresponding to Fe_3O_4 in the XRD curve confirms the successful formation of Fe_3O_4 during the modification process with MWCNT. This confirmation is essential because Fe_3O_4 is easily oxidized into maghamite which potentially impact its microwave absorption properties.[24]

Furthermore, the formation of Fe_3O_4/MWCNT/ZnO was confirmed with a high peak assigned to ZnO according to ICSD-31052, whereas the small peak of Fe_3O_4/MWCNT still appears in this XRD pattern, confirming the formation of Fe_3O_4/MWCNT/ZnO. Furthermore, the successful confirmation of the formation of PANI-ES was achieved with the presence of a prominent peak at 2θ of approximately 25.34°, which is consistent with previous findings.[25] Moreover, the XRD pattern of Fe_3O_4/MWCNT/ZnO/PANI exhibits a similar pattern to Fe_3O_4/MWCNT/ZnO. However, the peaks of Fe_3O_4/MWCNT are absent as the reducing of crystallinity.

due to the presence of PANI. The reduction in crystallinity observed in the presence of PANI can indeed be attributed to the low crystalline nature of PANI.[26] Consequently, properties such as strength and hardness can be affected by this reduction in crystallinity. Figure 1(b) shows the impact of different PANI compositions (FMZP1, FMZP2, FMZP3, FMZP4) on the XRD pattern of the nanocomposite. The addition of PANI to the nanocomposite reduces the XRD peak, indicating a decrease in the crystallinity of the nanocomposite.

The XRD peaks in a nanocomposite can indeed provide valuable information about the formation of its components and also offer insights into the crystallite size of the corresponding materials. The Scherrer formula, as depicted in the following equations, is commonly used to estimate the crystallite size based on XRD data:[27,28]

$$D = \frac{K\lambda}{\beta\cos\theta} \quad (1)$$

where D represents the average crystallite size, K is the Scherrer constant (typically around 0.9), λ is the wavelength of the X-rays used, β is the full width at half maximum (FWHM) of the diffraction peak, and θ is the Bragg angle. Our analysis using equation (1) reveals that the crystallite size of Fe_3O_4 ranges from 10.57 nm to 32.02 nm. Similarly, the crystallite size of ZnO falls between 18.86 nm and 36.10 nm.

Furthermore, FTIR spectroscopy was conducted to obtain evidence of the nanocomposite's formation. Figure 2(a) shows the FTIR spectra of all the samples. The peaks observed at approximately 440, 580, and 620 cm^{-1} in the Fe_3O_4/MWCNT nanocomposite are attributed to the stretching vibration of Fe–O bonds,[29] indicating the presence of Fe_3O_4 in the composite. Additionally, there are distinct peaks in the Fe_3O_4/MWCNT nanocomposite observed between 700–1300, 1620–1680, 1475–1550, and 1410–1420 cm^{-1}. These peaks can be

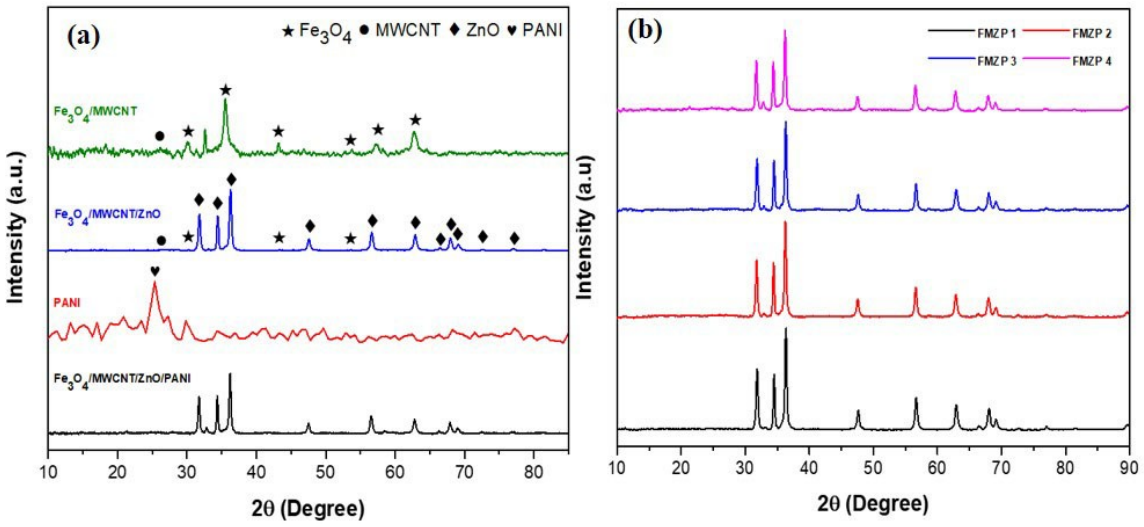


Figure 1. (a) XRD patterns of all samples; (b) XRD patterns of the nanocomposite with different compositions of polyaniline.

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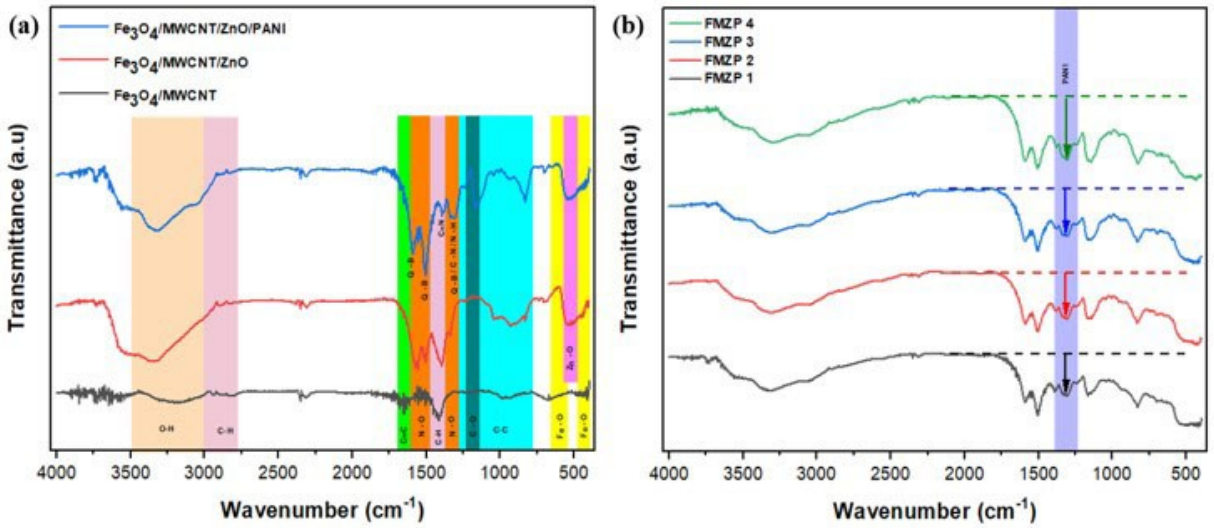


Figure 2. (a) FTIR spectra of all samples; (b) FTIR spectra of the nanocomposite with different compositions of polyaniline.

attributed to the vibration of C–C bonds, stretching of C=C bonds, asymmetric stretching of N–O bonds, and stretching of C–H bonds, respectively.[30] The observed bonds indicate the presence of MWCNT in the Fe_3O_4/MWCNT nanocomposite after functionalization with nitric acid. Additionally, the small peak detected within the range of 1150–1200 cm^{-1} can be attributed to the vibration of the C–O bond in the carboxyl (–COOH) group. This presence of the peak suggests that it may have occurred due to the oxidation of MWCNT during the functionalization. Furthermore, a prominent peak at approximately 457 and 545 cm^{-1} in the Fe_3O_4/MWCNT/ZnO nanocomposite suggests the presence of ZnO in the nanocomposite.[31] This finding further confirms the formation of the Fe_3O_4/MWCNT/ZnO nanocomposite. Moreover, the FTIR spectra of the Fe_3O_4/MWCNT/ZnO/PANI nanocomposite show noticeable peaks in the range of 1304–1310 cm^{-1}, which can be attributed to N–H bending and C–N stretching. A distinct peak is also observed at 1385 cm^{-1}, which can be attributed to $C{=}N^+$ stretching near the quinoid structure. Three peaks at 1308, 1508, and 1595 cm^{-1} indicate the presence of ring-stretching vibrations of the quinoid and benzenoid rings (Q–B).[32] These bonds confirm the existence of PANI in the Fe_3O_4/MWCNT/ZnO/PANI nanocomposite. Figure 3(b) shows the FTIR spectra of the Fe_3O_4/MWCNT/ZnO/PANI nanocomposite, illustrating different compositions of PANI. It is worth noting that within the highlighted region of 1304–1310 cm^{-1}, there are distinct peaks attributed to the N–H bending and C–N stretching of PANI. The intensity of these peaks increases as the composition of PANI increases from FMZP1 to FMZP4. This linear correlation provides further evidence of the formation of the Fe_3O_4/MWCNT/ZnO/PANI nanocomposite with an elevated PANI composition.

SEM-EDX analysis was conducted to examine the morphology of the Fe_3O_4/MWCNT/ZnO/PANI nanocomposite. Figure 3(a) shows the SEM image of the nanocomposite at magnifications of 50kx. The observation of nanotubes, as highlighted by a yellow circle, indicates the presence of MWCNTs in the nanocomposite. Additionally, the formation of a flake-like structure in the nanocomposite, as highlighted by a red circle, indicates the presence of PANI. This flake-like structure is consistent with previous results associating it with PANI.[33,34] The aggregation of nanoparticles may also be attributed to the formation of Fe_3O_4 and ZnO, as evidenced by the presence of Fe, Zn, and O in the nanocomposite according to the EDX curve as show in Figure 3(b). Furthermore, the composition analysis from EDX indicates that the nanocomposite consists of 3.92% Fe, 47.43% Zn, and 48.65% O. Interestingly, assuming that 3.92% of Fe and 5.23% of O (Fe:O = 3:4) were used in the formation of Fe_3O_4, it can be inferred that the remaining 43.42% of O and 47.43% of Zn (Zn:O = 1:0.92) contributed to the formation of ZnO in the nanocomposite. This evidence further confirms the formation of the Fe_3O_4/MWCNT/ZnO/PANI nanocomposite.

To confirm that the aggregation of nanoparticles could be attributed to the formation of ZnO and Fe_3O_4, we conducted a further analysis of the spherical particle size distribution in the nanocomposite morphology. Figure 3(c) illustrates the magnification of the spherical particles that were analyzed for their size. Our analysis revealed an average spherical particle size of approximately 26.77 nm, with a standard deviation of 4.82 nm, as depicted in Figure 3(d). Notably, this size range aligns with the crystallite size range of ZnO and Fe_3O_4 determined by the Scherrer formula. Although this finding does not directly confirm the formation of ZnO and Fe_3O_4, it provides additional support to the previous information obtained from XRD, EDAX, and FTIR analyses.

Optical properties of Fe_3O_4/MWCNT/ZnO/PANI nanocomposite

Figure 4(a) shows the Ultraviolet-visible (UV-Vis) spectral characteristics of the samples. The absorbance peak of Fe_3O_4/MWCNT was observed at 428.5 nm, which is similar to the previously

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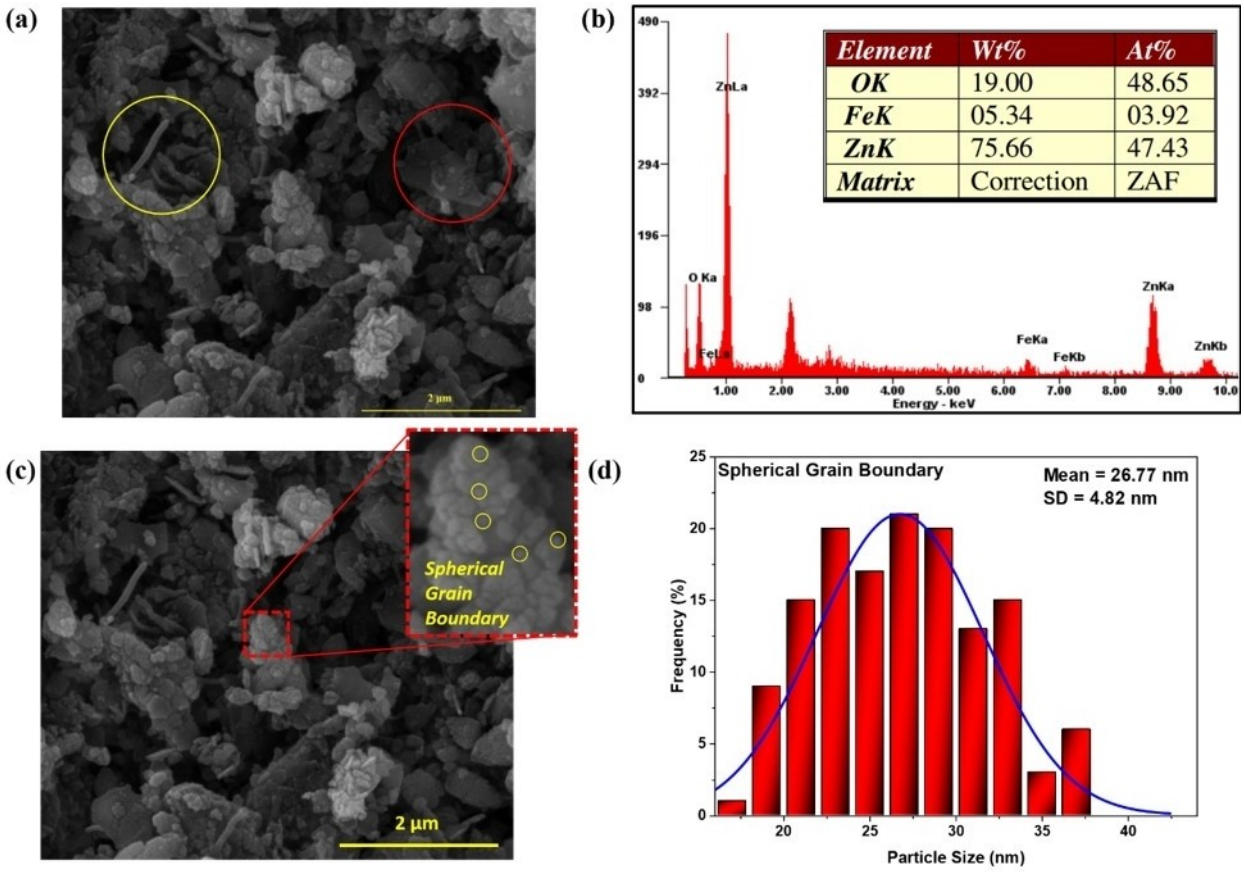


Figure 3. (a) SEM image, (b) elemental composition of Fe_3O_4/MWCNT/ZnO/PANI nanocomposite, (c) Magnification of the spherical grain boundary in the morphology of the nanocomposite, (d) Size distribution of spherical particles in the morphology of the nanocomposite.

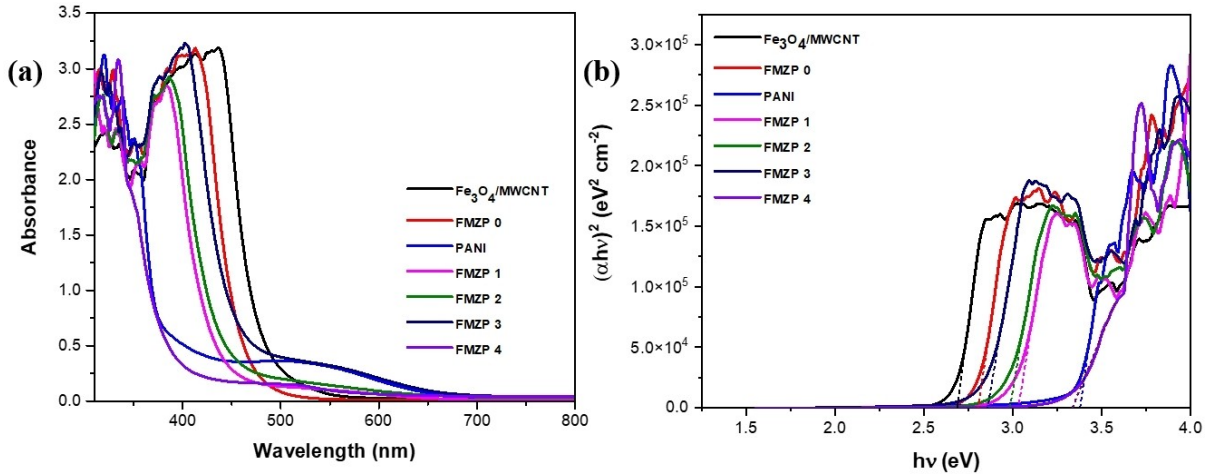


Figure 4. (a) UV-Vis spectral and (b) Tauc plot of all samples.

reported absorbance peak of Fe_3O_4.[35,36] This absorbance peak can be attributed to the surface plasma resonance band in the Fe_3O_4 nanoparticle.[37] This evidence further confirms the formation of Fe_3O_4 in the nanocomposite.

However, the absorbance peak of Fe_3O_4/MWCNT/ZnO is slightly shifted to the left because of the presence of ZnO, with an absorbance peak at approximately 377 nm through a band-gap absorption mechanism.[38] Moreover, PANI exhibited a prominent peak in the range 300–380 nm, indicating the π–π* transition of PANI. Another peak was observed in the range 450–600 nm, corresponding to both the polaron and bipolaron transitions associated with the excitation from B to Q ring structures in PANI.[39] Particularly, Fe_3O_4/MWCNT/ZnO/PANI exhibited three main peaks; two peaks because of PANI and one

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peak because of Fe_3O_4/MWCNT/ZnO. This further confirms the presence of all three components in the nanocomposite.

Furthermore, the optical band-gap properties of all samples were determined using the Tauc plot, which is shown in Figure 4(b). The direct optical band-gap energy can be estimated using equation (2):[40]

$$(\alpha h\nu)^2 = \beta(h\nu - E_g) \quad (2)$$

where α represents the absorption coefficient that depends on the absorbance and the thickness of the sample (as depicted in equation (3)); β is the constant of the equation; $h\nu$ is photon energy from UV-Vis; E_g is the direct band gap of the sample. The absorption coefficient α is calculated by using equation (3):

$$\alpha = \frac{2.303A}{d} \quad (3)$$

where A is the absorbance and d, are the thickness of the sample.

The direct band gap of all samples can be determined by extrapolating the linear region of the Tauc plot which involves finding the intercept between the extrapolated line and $h\nu$ axes (or $\alpha h\nu = 0$). Table 1 systematically presents the direct band gap values of all samples. For Fe_3O_4/MWCNT, the direct band gap was found to be 2.69 eV, which is lower than the reported direct band gap of 3.9 eV for pure Fe_3O_4 prepared using ammonia solution precipitation.[41] The presence of MWCNT in Fe_3O_4/MWCNT contributes to this lower band gap, as MWCNT has been shown to have a band gap-reducing effect on composite materials.[42,43] Furthermore, the band gap of FMZP0, a combination of Fe_3O_4/MWCNT/ZnO, was measured to be 2.80 eV. This value is slightly higher than the band gap of Fe_3O_4/MWCNT but lower than that of pure ZnO nanoparticles, which is approximately 3.26 eV.[44] This evidence can be attributed to the effective medium theory (EMT), as illustrated in equation (4):[45]

$$(1-\phi)\frac{\lambda_1 - \lambda_e}{\lambda_1 + 2\lambda_e} + \phi\frac{\lambda_2 - \lambda_e}{\lambda_2 + 2\lambda_e} = 0 \quad (4)$$

In equation (4), λ_1 represents the physical property of the continuous phase in the nanocomposite; λ_2 represents the physical property of the dispersed phase in the nanocomposite; λ_e represents the effective property of the nanocomposite; ϕ represents volume fraction of the dispersed phase. According to equation (4), the physical properties of the nanocomposite should lie within the range of physical properties exhibited by its components.

Furthermore, the Tauc plot analysis of PANI revealed that its direct optical band gap was found to be 3.38 eV, which is close to the reported band gap of conductive PANI at 3.2 eV. Following the effective medium theory, the direct optical band gap of the Fe_3O_4/MWCNT/ZnO/PANI nanocomposite is expected to theoretically range between 2.8 eV (the direct optical band gap of Fe_3O_4/MWCNT/ZnO) and 3.38 eV (the direct optical band gap of PANI). This estimation has been confirmed by Tauc plot analysis for the nanocomposite, which revealed that the direct optical band gaps of FMZP1, FMZP2, FMZP3, and FMZP4 were found to be 3.03, 2.98, 2.86, and 3.34 eV, respectively, as recorded in Table 1. However, it is evident that the variation in the direct optical band gap of the nanocomposite is non-linear with increasing PANI substitution. This non-linear variation can be attributed to several factors, including intermolecular interactions,[46] morphological changes,[47] and aggregation effect.[48] These factors collectively contribute to alterations in the electronic structure of the nanocomposite, resulting in the observed non-linear change in the optical band gap.

Table 1. The direct optical band gap of all samples.

Sample	Direct Band Gap (eV)
Fe_3O_4/MWCNT	2.69
Fe_3O_4/MWCNT/ZnO (FMZP0)	2.80
PANI	3.38
FMZP1	3.03
FMZP2	2.98
FMZP3	2.86
FMZP4	3.34

Microwave absorption properties of Fe_3O_4/MWCNT/ZnO/PANI nanocomposite

The microwave absorption properties of all samples were examined using a Vector Network Analyzer (VNA), with a RL falling within the range of 8–12 GHz at room temperature. Figure 5 shows the microwave RL curve of all samples with a consistent thickness of 1 mm. It is evident that FMZP0 or Fe_3O_4/MWCNT/ZnO exhibits the highest RL peak of −10.74 dB with a narrow EB (at RL < −10 dB) of 0.64 GHz. As previously mentioned in the introduction, an RL of less than −10 dB corresponds to more than 90% electromagnetic wave absorption, while a high RL suggests that the material is not effective

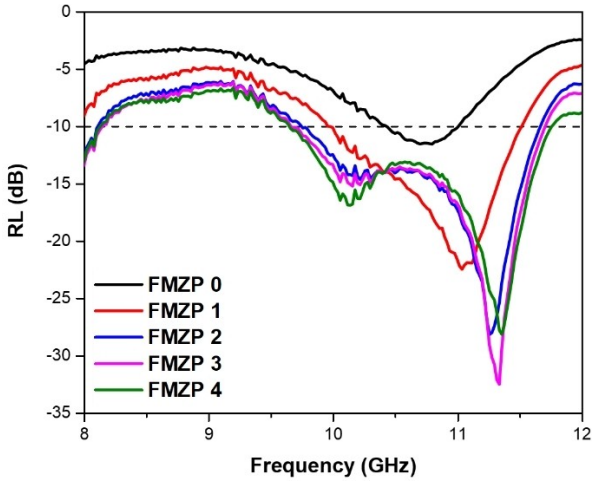


Figure 5. Reflection loss curve of all samples with a thickness of 1 mm.

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in absorbing electromagnetic waves. Furthermore, a narrow EB indicates poor absorption properties of the material for long-range electromagnetic microwaves. Therefore, enhancing the dielectric loss properties could be a crucial factor in improving microwave absorption performance.[49]

In this study, we introduced a conductive polymer, PANI, as a component in the nanocomposite to address the absence of the dielectric loss component. As a result, the Fe_3O_4/MWCNT/ZnO/PANI nanocomposite with different compositions of PANI exhibited two split RL peaks below −10 dB. This splitting effect increases the EB of microwave absorption. Specifically, FMZP1 exhibited a peak RL of −22.43 dB with a corresponding EB of 1.57 GHz, demonstrating an improvement compared to FMZP0. This highlights the role of PANI as an additional component to compensate for the lack of dielectric loss in the nanocomposite. Further addition of PANI enhances the microwave absorption properties of the nanocomposite. FMZP2, FMZP3, and FMZP4 exhibited minimum RL values of −28.15 dB (with an EB of 2.05 GHz), −32.66 dB (with an EB of 2.15 GHz), and −28.03 dB (with an EB of 2.26 GHz), respectively. The presence of PANI, a conductive polymer, clearly has a significant impact on tuning the microwave absorption properties of the nanocomposite.

Moreover, when considering reflection loss, it is clear that FMZP3 shows the highest peak, signifying the optimal reflection loss among all the samples. This optimal condition is attained by increasing the amount of PANI in the nanocomposite, which enhances its conductivity while causing a decrease inother properties. As a result, it is possible that after reaching the optimum condition, the main peak of reflection loss reduces, whileother properties like electromagnetic bandwidth may increase.

In principle, the tunable microwave absorption properties of the nanocomposite can be attributed to all components of the microwave absorber, encompassing the magnetic properties of Fe_3O_4, dielectric properties of ZnO, and conductive properties of MWCNT and PANI. However, in this study, PANI, which is an easily prepared compound, has demonstrated remarkable tunable properties in microwave absorption.

Through a comparative analysis, our newly developed nanocomposite, despite having a small thickness, surpasses previously reported materials in terms of either RL peak or EB, as indicated in Table 2.

A modified theoretical model on microwave absorption properties of Fe_3O_4/MWCNT/ZnO/PANI nanocomposite

To further investigate the role of PANI in enhancing nanocomposite performance, a modified theoretical model was introduced. This approach involved extracting RL data to

determine the nanocomposite's complex relative permittivity and permeability. Generally, the material responds to the external magnetic and external electric fields, as shown in Figure 6(a). Accordingly, the transmission line theory accommodates both responses in terms of RL, as depicted in the following equation:[66]

$$RL = -20\ \mathrm{Log}\left(\frac{\sqrt{\left|\frac{\mu_r}{\varepsilon_r}\right|}\ -\ 1}{\sqrt{\left|\frac{\mu_r}{\varepsilon_r}\right|}\ +\ 1}\right) \tag{5}$$

where RL is the reflection loss of the material; μ_r is the complex relative permeability of the material representing the magnetic response of the material; ε_r is the complex relative permittivity

Table 2. Microwave absorption properties comparison between some previously reported material and this development.

Material	Thickness (mm)	RL peak (dB)	Bandwidth (GHz) at RL < −10dB	Ref.
Ni@Graphite	1.78	−19.27	3.2	[50]
$FeCo_3P/Al_2O_3$	2.6	−26.15	4.2	[51]
FeOx/CSBC	2.5	−26.5	6.5	[52]
Fe_3O_4 NPs	6	−21.12	3.4	[53]
Fe_3O_4 NPs/Paraffin wax	4	−47.8	0.88	[54]
Graphene@Fe_3O_4@PANI@WO_3	4	−46.7	1.8	[55]
Fe_3O_4/SiO_2/graphene	1.5	−27.1	3.3	[56]
Fe_3O_4/rGO	3	−22.7	3.13	[57]
$CoFe_2O_4$ NPs	2.5	−5	–	[58]
$CoFe_2O_4$ hollow sphere/graphene	2	−18.5	3.9	[59]
Mesoporous $CoFe_2O_4$	2	−18	4.5	[60]
$CoFe_2O_4/Co_3Fe_7$-Co core/shell	4	−34.4	1	[61]
SiC	3	−17.4	2.5	[62]
Fe_3O_4/Ppy/CNT	3	−25.9	4.8	[63]
Fe_3O_4/ZnO	3.5	−22.7	5.9	[64]
RGO/$MnFe_2O_4$ + PVDF	3	−29	4.88	[65]
Fe_3O_4/MWCNT/ZnO/PANI	1	−32.66	2.15	**This Work**

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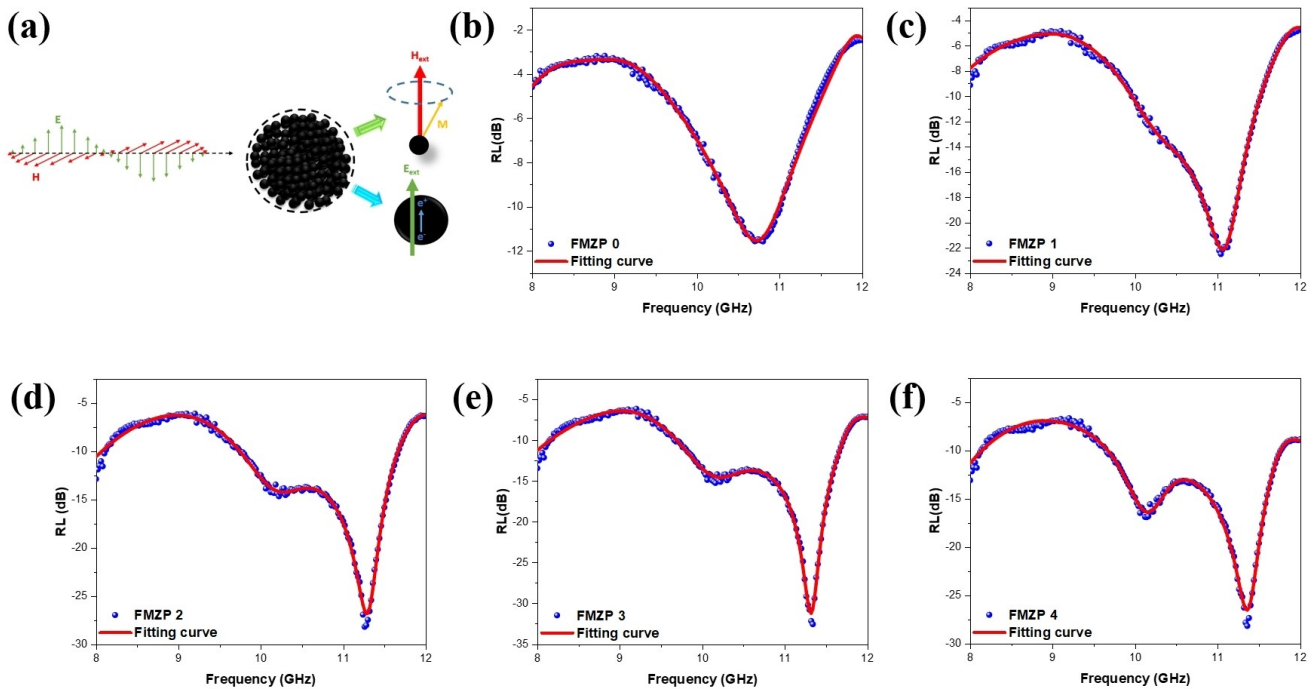


Figure 6. (a) The scheme of the material response toward the incident electromagnetic wave; the reflection loss curve and its fitting for (b) FMZP0; (c) FMZP1; (d) FMZP2; (e) FMZP3; (f) FMZP4.

of the material representing the electric response of the material.

The complex relative permeability of the material was manifested by the propagation of magnetic moment due to the presence of an external magnetic field from electromagnetic waves. Landau-lifshifts theory describes the magnetic response of the material in terms of complex relative permeability, as shown in the following equation:[67]

$$\mu_r = 1 + 4\pi \sum_j \frac{{\omega_j}^2 + i\omega\gamma_j}{\beta({\omega_j}^2 - \omega^2 + 2i\omega\gamma_j)} \tag{6}$$

ω is angular frequency of the electromagnetic wave, $\beta = \frac{1}{kT}$, $\omega_j = \mu_0 \beta s_j$; γ_j is the damping constant. Additionally, s_j in the definition of ω_j is the total magnetic moment of a domain in the material.

Furthermore, the complex relative permittivity of the material can be found using the Drude–Lorentz theory. According to this theory, the complex relative permittivity of the material can be expressed as follows:[68]

$$\varepsilon_r = 1 + \frac{Nq^2}{m\varepsilon_0} \sum_j \frac{f_j}{({\omega_j}^2 - \omega^2 + 2i\omega\gamma_j)} \tag{7}$$

N represents the number of molecules per unit volume; q represents the charge of an electric dipole; ω_j represents the angular frequency associated with the oscillation or vibration of the electric dipole; f_j represents the number of electrons with angular frequency of ω_j; the damping constant γ_j corresponds to the rate at which the energy of a system with angular frequency ω_j decreases due to damping; m denotes mass of electron; ε_0 is the permittivity of vacuum.

By incorporating equation 5–7 analytically, the new representation model of the reflection can be determined as follows:

$$RL = -20\ \mathrm{Log}\left|\frac{\sqrt{\left|\frac{\mu_{re}}{\varepsilon_{re}}\right|}\sqrt{\frac{\sqrt{\left(1+\frac{\mu_{im}\varepsilon_{im}}{\mu_{re}\varepsilon_{re}}\right)^2+\left(\frac{\mu_{im}}{\mu_{re}}-\frac{\varepsilon_{im}}{\varepsilon_{re}}\right)^2}}{1+\left(\frac{\varepsilon_{im}}{\varepsilon_{re}}\right)^2}} - 1}{\sqrt{\left|\frac{\mu_{re}}{\varepsilon_r e}\right|}\sqrt{\frac{\sqrt{\left(1+\frac{\mu_{im}\varepsilon_{im}}{\mu_{re}\varepsilon_{re}}\right)^2+\left(\frac{\mu_{im}}{\mu_{re}}-\frac{\varepsilon_{im}}{\varepsilon_{re}}\right)^2}}{1+\left(\frac{\varepsilon_{im}}{\varepsilon_{re}}\right)^2}} + 1}\right| \tag{8}$$

where

$$\varepsilon_{re} = 1 + \sum_i \frac{\frac{Nq^2 f_i}{m\varepsilon_0}({\omega_i}^2 - \omega^2)}{({\omega_i}^2 - \omega^2)^2 + {\gamma_i}^2\omega^2} \tag{9}$$

$$\varepsilon_{im} = \sum_i \frac{2Nq^2 f_i \gamma_j \omega / m\varepsilon_0}{({\omega_i}^2 - \omega^2)^2 + {\gamma_i}^2\omega^2} \tag{10}$$

$$\mu_{re} = 1 + \sum_i \frac{\frac{4\pi{\omega_i}^2}{\beta}\left({\omega_i}^2 - \omega^2 + 2{\gamma_i}^2\left(\frac{\omega}{\omega_i}\right)^2\right)}{({\omega_i}^2 - \omega^2)^2 + 4{\gamma_i}^2\omega^2} \tag{11}$$

$$\mu_{im} = \sum_i \frac{\frac{8\pi{\omega_i}^2\gamma_i\omega}{\beta}\left(1 + \frac{\left[{\omega_i}^2 - \omega^2\right]}{2{\omega_i}^2}\right)}{({\omega_i}^2 - \omega^2)^2 + 4{\gamma_i}^2\omega^2} \tag{12}$$

Using this modified model, the reflection loss curves of all samples were excellently fitted, as shown in Figure 6(b)–(f).The

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parameters obtained from the fitting were then used to calculate the complex relative permittivity and permeability of all samples. Figure 7(a)–(d) visually represent the complex relative permittivity and permeability using Equation (8) and the obtained parameters.

Furthermore, as shown in Figure 7(b), it is evident that the inclusion of polyaniline significantly enhances the maximum value of the imaginary component of the complex relative permittivity in the nanocomposite. This increase in the peak of the imaginary component is typically associated with increased material conductivity, which has been supported by previous studies.[69] Given the high conductivity of polyaniline, it is reasonable to conclude that its integration into the nanocomposite would enhance the overall conductivity of the material, thereby leading to the observed rise in the peak of the imaginary part of the complex relative permittivity.

In addition, the rise in the peak of the imaginary part of the complex relative permittivity is observed at a frequency of approximately 10.2 GHz, aligning with the secondary peak detected in the reflection loss curve. This suggests that the inclusion of polyaniline causes an elevation in the peak of the imaginary part of the complex relative permittivity, which subsequently leads to an increase in the secondary peak of the reflection loss. This enlargement of the secondary peak is directly linked to the broadening of the electromagnetic bandwidth of the nanocomposite.

An interesting phenomenon occurs in the reflection loss of the nanocomposite after the addition of polyaniline, as discussed in the previous section. It has been observed that there is an optimal condition in which the reflection loss of the nanocomposite is reduced. According to equation (5), the reflection loss tends to be higher when the magnitudes of the complex relative permittivity and permeability are closer to eachother. To investigate the optimal condition of FMZP3 and address the observed decrease in reflection loss in the last variation, FMZP4, we analyzed the complex relative permittivity and permeability of both samples at their respective reflection loss peak frequencies of 11.33 GHz and 11.36 GHz.

Our analysis, using simulated complex relative permittivity and permeability as shown in Figure 7(a–d), revealed that the magnitudes of the complex relative permittivity and permeability for FMZP3 were 0.26 and 0.28, respectively. In contrast, for FMZP4, the magnitudes were 0.33 and 0.39, respectively. The closer proximity of the complex relative permittivity and permeability values in FMZP3, compared to FMZP4, resulted in a higher reflection loss, which is consistent with the expectations of the transmission line theory. Therefore, the observed decrease in reflection loss in the last variation can be attributed to the closer magnitudes of the complex relative permittivity and permeability in FMZP3, compared to FMZP4.

As a summary of simulation, the inclusion of polyaniline results in a noticeable augmentation in a smaller peak within

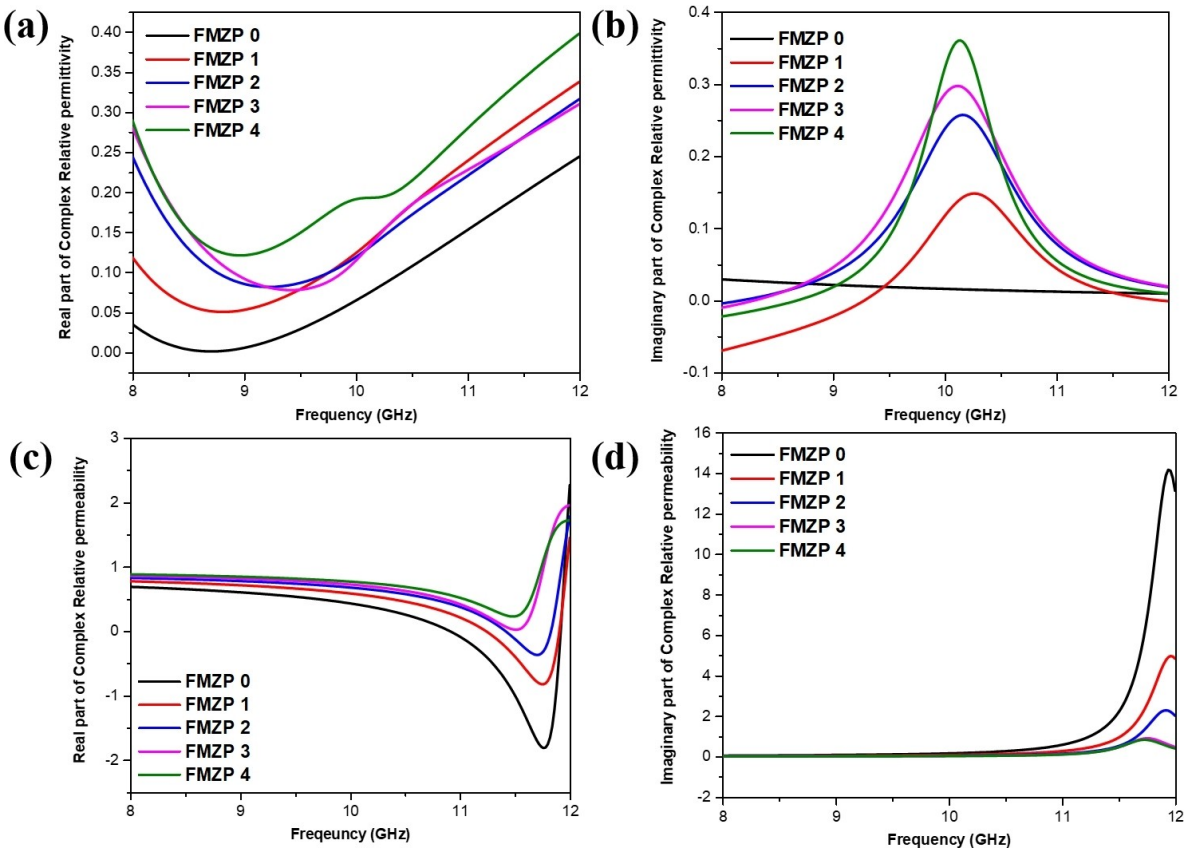


Figure 7. (a) Simulated real part of complex relative permittivity; (b) Simulated imaginary part of complex relative permittivity; (c) Simulated real part of complex relative permeability; (d) Simulated imaginary part of complex relative permittivity of all samples.

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the reflection loss curve at approximately 10 GHz. This enhancement is associated with an increase in the peak value of the imaginary part of the complex permittivity and a concurrent decrease in the peak value of the imaginary part of the complex permeability. These changes in the peak values for both the imaginary part of the complex permittivity and permeability serve to expand the overall electromagnetic bandwidth of the nanocomposite.

Conclusions

In this study, a nanocomposite consisting of Fe_3O_4/MWCNT/ZnO/PANI was successfully synthesized using a facile route preparation technique. The presence of all components in the nanocomposite was confirmed through XRD characterization, which revealed characteristic peaks. Increasing the PANI composition resulted in reduced crystallinity, attributed to the polymer matrix covering the crystalline structure. FTIR analysis further confirmed the presence of various vibration mechanisms in the nanocomposite, including Zn–O, Fe–O, C–N, C–C, C=C, and quinoid-benzenoid rings stretching. SEM-EDAX analysis provided insights into the morphology and distribution of components, showing an average spherical particle size of approximately 26.77 nm, falling within the range of Fe_3O_4 and ZnO crystallite sizes determined from XRD. The nanocomposite exhibited an optical band gap ranging from 2.69 to 3.38 eV, influenced by the PANI composition through possible mechanism such as intermolecular interactions, morphological changes, and charge transfer effects. Remarkably, the nanocomposite demonstrated satisfactory microwave absorption properties, with a minimum reflection loss (RL) of −32.66 dB for the composition containing 37.50% PANI by weight and a maximum electromagnetic bandwidth (EB) of 2.26 GHz for the composition containing 44.44% PANI by weight, indicating its potential as an effective shielding material. Additionally, a modified theoretical model combining transmission line, Landau-Lifshits, and Drude-Lorentz theories provided insights into the role of polyaniline in enhancing microwave absorption as measured by RL. However, the nanocomposite could exhibit instability under specific conditions, highlighting the need for stability enhancement in future studies. Nonetheless, the modified model presented in this study holds promise as an experimental guide for designing efficient microwave absorbers.

Experimental Section

Material

The materials used in this study included iron sand collected from Sine beach, Indonesia, chloric acid (HCl) (Merck, 12 M), ammonium hydroxide (NH_4OH) (Merck, 14 M), multi-walled carbon nanotubes (MWCNT) (Sigma Aldrich, approximately 50 nm in diameter and 1 μm in length), nitric acid (HNO_3) (Merck, 37%), zinc acetate dihydrate ($Zn(CH_3CO_2)_2{\bullet}2H_2O$) (Merck), sodium hydroxide (NaOH) (Merck, 3 M), aniline (Merck, 99%), ammonium persulfate ($(NH_4)_2S_2O_8$) (Merck, 99%), deionized water, ethanol, methanol, and distilled water. Prior to use, the iron sand underwent a separation process involving a permanent magnet.

Preparation of Fe_3O_4/MWCNT/ZnO and Fe_3O_4/MWCNT/ZnO/PANI nanocomposite

The MWCNT/Fe_3O_4 nanocomposite was prepared using the co-precipitation method. The preparation of functionalized MWCNT involved dissolving 1 g of MWCNT powder in nitric acid (1 : 100 w/v). The mixture was then sonicated at a frequency of 40 KHz for 120 min at a moderate temperature of 50 °C. After that, the solution was filtered using filter paper, and subsequently washed with ethanol and distilled water until it reached a neutral pH. Finally, the black precipitate was dried at 100 °C for 3 h, resulting in the functionalized MWCNT sample.

Iron powder was first dissolved in HCl and stirred at 900 rpm for 30 min. Then, 10 ml of the resulting solution was mixed with 0.1 g of MWCNT and sonicated at 40 KHz for 10 min. The solution underwent a titration process with 18 ml of NH_4OH for 20 min until a dark brown precipitate formed. The resulting precipitate was washed multiple times and dried at 100 °C for 3 h to obtain the MWCNT/Fe_3O_4 nanocomposite. Figure 8(a) illustrates the detailed process of the MWCNT/Fe_3O_4 preparation.

ZnO was also prepared using the co-precipitation method. First, 6.57 g of $Zn(CH_3CO_2)_2{\bullet}2H_2O$ was dissolved in 100 ml of deionized water and stirred at 720 rpm for 1 h. The solution underwent a titration process with NH_4OH until the pH reached 13. The solution was then covered with plastic and heated at 90 °C using a hot plate for 1 h. Subsequently, 250 ml of ethanol was added to the solution and left to age until a white precipitate formed at the bottom of the beaker glass. The resulting precipitate was washed several times and dried at 100 °C for 1 h to obtain ZnO. Figure 8(b) shows the detailed process of ZnO preparation.

PANI emeraldine salt (PANI-ES) was synthesized through oxidative polymerization, according to a previous study.[70] Aniline (2 ml) was dissolved in HCl (0.2 M, 50 ml) to form solution A. Simultaneously, ammonium persulfate (6.275 g) was dissolved in deionized water (50 ml) to form solution B. Both solutions were aged for 1 h. Afterward, they were combined in a glass beaker and stirred at 700 rpm for 2 h. The resulting mixture was left to age for 24 h until a dark green precipitate formed at the bottom of the beaker. Finally, the green precipitate was washed multiple times with HCl and dried using a hotplate at 40 °C to obtain PANI powder. Figure 8(c) shows the preparation process of PANI.

The MWCNT/Fe_3O_4/ZnO/PANI nanocomposite was prepared using an easily physical mixture strategy. The process involved dissolving 4 g of MWCNT/Fe_3O_4 nanocomposite in 50 ml of deionized water and sonicating it for 30 min at 40 KHz. Then, 16 g of ZnO were added to the sonicated solution and stirred at 720 rpm for approximately 2 h until the solution became homogenous in color. The resulting precipitate was washed several times and dried at 100 °C for 1 h to obtain the MWCNT/Fe_3O_4/ZnO nanocomposite. Finally, 1.5 g of MWCNT/Fe_3O_4/ZnO nanocomposite and different amounts (0.3 g (5 : 1), 0.6 g (5 : 2), 0.9 g (5 : 3), and 1.2 g (5 : 4)) of PANI were dissolved in 50 ml of deionized water and stirred at 700 rpm for 2 h. The resulting precipitate was washed multiple times and dried at 100 °C for 1 h to obtain the MWCNT/Fe_3O_4/ZnO/PANI nanocomposite. Figure 8(d) shows the detailed preparation process of MWCNT/Fe_3O_4/ZnO/PANI nanocomposite. The resulting MWCNT/Fe_3O_4/ZnO/PANI nanocomposite, with mass ratios of MWCNT/Fe_3O_4/ZnO to PANI (5 : 0, 5 : 1, 5 : 2, 5 : 3, 5 : 4), was

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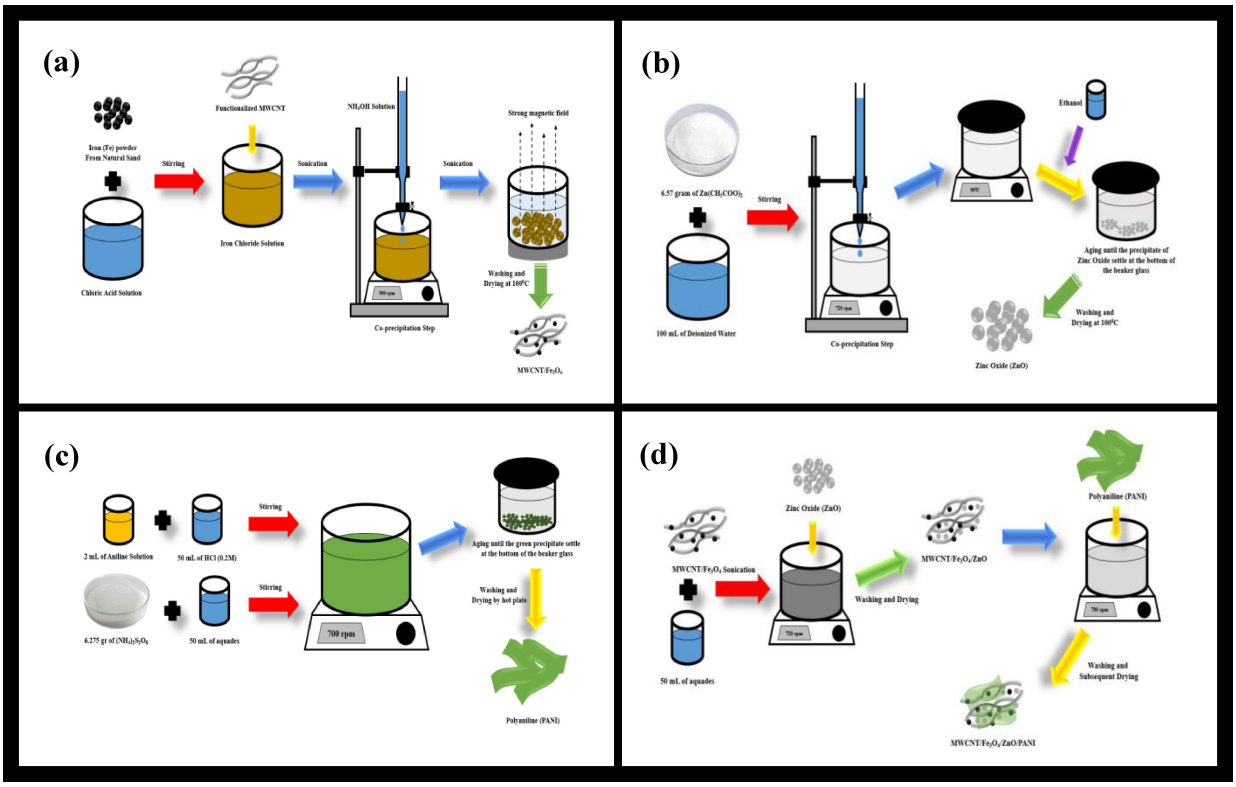


Figure 8. Preparation scheme of (a) Fe_3O_4/MWCNT, (b) ZnO, (c) polyaniline (PANI), (d) Fe_3O_4/MWCNT/ZnO/PANI

designated as FMZP0, FMZP1, FMZP2, FMZP3, and FMZP4, respectively.

Characterizations

The crystal structure of the samples was examined using X-ray diffraction (XRD) with an XPERT-PRO diffractometer, which generated CuKα radiation of 1.5406 Å at 40 kV and 35 mA. The transmittance of the samples within the infrared frequency range was measured using a Shimadzu infrared spectrometer through Fourier transform infrared spectroscopy (FTIR). The nanocomposite morphology and elemental content were examined using a scanning electron microscope with energy dispersive X-ray spectroscopy (SEM-EDX). The optical properties of the sample were examined using a Shimadzu ultraviolet (UV-1700) spectrophotometer. Finally, microwave absorption properties were measured using a vector network analyzer (VNA) with a frequency range of 8–12 GHz and a sample thickness of 1 mm.

Acknowledgements

This research was supported by internal research grant from Universitas Negeri Malang for AT.

Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords: Fe_3O_4/MWCNT/ZnO/PANI · nanocomposite · facile route synthesis · iron sand · microwave absorption · modified theoretical model

[1] J.-S. Li, H. Huang, Y.-J. Zhou, C.-Y. Zhang, Z.-T. Li, *J. Mater. Res.* **2017**, *32*, 1213–1230, DOI 10.1557/jmr.2017.80.
[2] H. Zhang, S. Lin, *Membranes.* **2023**, *13*, 315, DOI 10.3390/membranes13030315.
[3] H. Pang, Y. Duan, L. Huang, L. Song, J. Liu, T. Zhang, X. Yang, J. Liu, X. Ma, J. Di, X. Liu, *Compos. Part B Eng.* **2021**, *224*, 109173, https://doi.org/10.1016/j.compositesb.2021.109173.
[4] F. Meng, H. Wang, F. Huang, Y. Guo, Z. Wang, D. Hui, Z. Zhou, *Compos. Part B Eng.* **2018**, *137*, 260–277, DOI 10.1016/j.compositesb.2017.11.023.
[5] W. Aiqiong, J. Li, M. Chen, X. Zhao, *J. Ind. Text.* **2022**, *52*, 152808372211331, DOI 10.1177/15280837221133113.
[6] X. Zhao, Z. Zhang, L. Wang, K. Xi, Q. Cao, D. Wang, Y. Yang, Y. Du, *Sci. Rep.* **2013**, *3*, 3421, https://doi.org/10.1038/srep03421.
[7] L. B. Kong, L. Liu, Z. Yang, S. Li, T. Zhang, C. Wang, *In: Magnetic, Ferroelectric, and Multiferroic Metal Oxides.* **2018**, 361–385. Elsevier.
[8] I. Mohammed, J. Mohammed, A. K. Srivastava, *Cryst. Res. Technol.* **2023**, *58*, 2200200, https://doi.org/10.1002/crat.202200200.
[9] R. S. Yadav, I. Kuřitka, *Crit. Rev. Solid State Mater.* **2023**, Sci. 1–37, https://doi.org/10.1080/10408436.2023.2214577.
[10] Y. Wang, *Int. J. Mater. Res.* **2014**, *105*, 3–12, DOI 10.3139/146.110996.
[11] X. Su, Y. Liu, Z. Liao, Y. Bi, M. Ma, Y. Chen, Y. Ma, F. Wan, K. L. Chung, *Synth. Met.* **2022**, *291*, 117190, DOI 10.1016/j.synthmet.2022.117190.
[12] Yusmaniar, Erdawati, H. Sosiati, S. Budi, M. Alaydrus, E. Handoko, *Mater. Res. Express.* **2021**, *8*, 046101, https://doi.org/10.1088/2053-1591/abf52a.

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[13] H. Gupta, R. Khanna, M. K. Rai, S. B. Narang, *Waste Biomass Valorization.* **2023**, 1–11 https://doi.org/10.1007/s12649-023-02282-6.
[14] M. A. Kholil, P. Priyono, A. Subagio, *Mater. Res. Express.* **2022**, *9*, 025003, https://doi.org/10.1088/2053-1591/ac4e3a.
[15] N. Yulfriska, Z. Affandi, Yohandri, L. Dwiridal, R. Ramli, *J. Phys. Conf. Ser.* **2020**, *1481*, 012006, https://doi.org/10.1088/1742-6596/1481/1/012006.
[16] H. Hekmatara, M. Seifi, K. . Forooraghi, *J. Magn. Magn. Mater.* **2013**, *346*, 186–191, https://doi.org/10.1016/j.jmmm.2013.06.032.
[17] X. Sun, G. Ma, X. Lv, M. Sui, H. Li, F. Wu, Wang, *J. Materials.* **2018**, *11*, 780, https://doi.org/10.3390/ma11050780.
[18] A. Taufiq, S. T. U. Intan Subadra, N. Hidayat, S. Sunaryono, A. Hidayat, E. Handoko, M. Munasir, M. Alaydrus, L. Chuenchom, *Int. J. Nanosci. Nanotechnol.* **2021**, *17*, 41–53.
[19] Y. Wang, X. Gao, Y. Fu, X. Wu, Q. Wang, W. Zhang, C. Luo, *Compos. Part B Eng.* **2019**, ***169***, 221–228.
[20] K. Praveena, M. . Bououdina, *J. Electron. Mater.* **2020**, *49(10)*, 6187–6198.
[21] R. V. Lakshmi, P. Bera, R. P. S. Chakradhar, B. Choudhury, S. P. Pawar, S. Bose, R. U. Nair, H. C. Barshilia, *Phys. Chem. Chem. Phys.* **2019**, *21(9)*, 5068–5077.
[22] X. Ding, G. K. Vegesna, H. Meng, A. Winter, B. P. Lee, *Chem. Phys.* **2015**, *216*, 1109–1119, https://doi.org/10.1002/macp.201500010.
[23] P. Nie, C. Min, H.-J. Song, X. Chen, Z. Zhang, K. Zhao, *Tribol. Lett.* **2015**, *58*, 7, https://doi.org/10.1007/s11249-015-0476-7.
[24] J. X. J. Zhang, K. Hoshino, *Molecular Sensors and Nanodevices* **2019**, 413–487.
[25] K T, V., Belagali, S.L, *IOSR J. Appl. Chem.* **2015**, *08*, 53–56, https://doi.org/10.9790/5736–0801025356.
[26] K. Rajakumar, S. D. Kirupha, S. Sivanesan, R. L. Sai, *J. Nanosci. Nanotechnol.* **2014**, *14(4)*, 2937–2946.
[27] S. Moradi, H. H. Ardeshiri, A. Gholami, H. Ghafuri, *Heliyon* **2023**, *9(11)*, e22185.
[28] M. Aghmasheh, M. A. Rezvani, V. Jafarian, H. H. Ardeshiri, *Energy Fuels* **2023**, *37(13)*, 9474–9486.
[29] S. Fekri Aval, A. Akbarzadeh, M. R. Yamchi, F. Zarghami, K. Nejati-Koshki, N. Zarghami, Artif, *Cells Nanomedicine Biotechnol.* **2016**, *44*, 188–193, https://doi.org/10.3109/21691401.2014.934456.
[30] A. B. D. Nandiyanto, R. Oktiani, R. Ragadhita, *J. Sci. Technol.* **2019**, *4*, 97, https://doi.org/10.17509/ijost.v4i1.15806.
[31] K. Handore, S. Bhavsar, A. Horne, P. Chhattise, K. Mohite, J. Ambekar, N. Pande, V. Chabukswar, *J. Macromol. Sci. Part A.* **2014**, *51*, 941–947, https://doi.org/10.1080/10601325.2014.967078.
[32] K. A. Ibrahim, *Arab. J. Chem.* **2017**, *10*, S2668–S2674, DOI 10.1016/j.arabjc.2013.10.010.
[33] S. Golba, M. Popczyk, S. Miga, J. Jurek-Suliga, M. Zubko, J. Kubisztal, K. Balin, *Materials.* **2020**, *13*, 5108, DOI 10.3390/ma13225108.
[34] J. Chen, X. Hong, Y. Zhao, Y. Xia, D. Li, Q. Zhang, *J. Mater. Sci.* **2013**, *48*, 7708–7717, https://doi.org/10.1007/s10853-013-7591-3.
[35] A. O. Kalu, E. C. Egwim, A. A. Jigam, H. L. Muhammed, *Nano Express.* **2022**, *3*, 045004, https://doi.org/10.1088/2632-959X/aca925.
[36] M. Mahdavi, F. Namvar, M. Ahmad, R. Mohamad, *Molecules.* **2013**, *18*, 5954–5964, https://doi.org/10.3390/molecules18055954.
[37] K. Rajendran, G. S. Balakrishnan, Kalirajan, *J. Int. J. PharmTech Res.* **2015**, *8*, 670–677.
[38] Z. M. AL-Asady, A. H. AL-Hamdani, M. A. Hussein, *2ND INTERNATIONAL CONFERENCE ON MATERIALS ENGINEERING & SCIENCE (IConMEAS 2019)*, Baghdad, Iraq, **2020**.
[39] A. Ibrahim, M. H. Abdel-Aziz, M. Sh. Zoromba, A. F. Al-Hossainy, *Synth. Met.* **2018**, *238*, 1–13, https://doi.org/10.1016/j.synthmet.2018.02.006.
[40] J. Tauc, R. Grigorovici, A. Vancu, *Phys. Status Solidi B.* **1966**, *15*, 627–637, https://doi.org/10.1002/pssb.19660150224.
[41] E. C. Nnadozie, P. A. Ajibade, *Int. J. Electrochem. Sci.* **2022**, *17*, 22124, https://doi.org/10.20964/2022.12.05.
[42] K. F. Chan, M. H. Mohd Zaid, S. Liza, M. S. Mamat, K. A. Matori, N. A. Endot, M. Tanemura, Y. Yaakob, *Ceram. Int.* **2021**, *47*, 20108–20116, https://doi.org/10.1016/j.ceramint.2021.04.017.
[43] H. Kh. Rasheed, A. A. Kareem, *J. Opt. Commun.* **2021**, *42*, 25–29, https://doi.org/10.1515/joc-2018-0024.
[44] S. Agarwal, L. K. Jangir, K. S. Rathore, M. Kumar, K. Awasthi, *Appl. Phys. A.* **2019**, *125*, 553, https://doi.org/10.1007/s00339-019-2852-x.
[45] M. Wang, N. Pan, *Mater. Sci. Eng. R Rep.* **2008**, *63*, 1–30, https://doi.org/10.1016/j.mser.2008.07.001.
[46] M. Filapek, M. Matussek, A. Szlapa, S. Kula, M. Pająk, *Electrochimica Acta* **2016**, *216*, 449–456.
[47] Y. Dong, M. Kapilashrami, Y. Zhang, J. Guo, *CrystEngComm* **2013**, *15(48)*, 10657.
[48] Y. Ru, G. I. N. Waterhouse, S. Lu, *Aggregate* **2022**, *3(6)*, e296.
[49] Y. Wang, D. Sun, G. Liu, W. Jiang, *Adv. Powder Technol.* **2015**, *26*, 1537–1543, https://doi.org/10.1016/j.apt.2015.08.014.
[50] J. Yang, Y. Liu, J. Si, L. Lu, J. Xu, X. Su, *Synth. Met.* **2023**, *293*, 117255, https://doi.org/10.1016/j.synthmet.2022.117255.
[51] L. Zhou, X. Liu, F. Luo, J. Yu, D. Zhang, Z. Wang, H. Jia, H. Wang, *Ceram. Int.* **2023**, *49*, 30417–30423, DOI 10.1016/j.ceramint.2023.06.304.
[52] L. Wang, H. Guan, S. Su, J. Hu, Y. Wang, *J. Alloys Compd.* **2022**, *903*, 163894, https://doi.org/10.1016/j.jallcom.2022.163894.
[53] G. Wang, Y. Chang, L. Wang, L. Liu, C. Liu, *Mater. Res. Bull.* **2013**, *48*, 1007–1012, https://doi.org/10.1016/j.materresbull.2012.11.089.
[54] L. Zhang, P. Dai, X. Yu, Y. Li, Z. Bao, J. Zhu, K. Zhu, M. Wu, X. Liu, G. Li, H. Bi, *Appl. Surf. Sci.* **2015**, *359*, 723–728, https://doi.org/10.1016/j.apsusc.2015.10.163.
[55] Y. Wang, X. Wu, W. Zhang, C. Luo, J. Li, Q. Wang, *Synth. Met.* **2017**, *231*, 7–14, https://doi.org/10.1016/j.synthmet.2017.06.013.
[56] X. Liu, Y. Chen, C. Hao, J. Ye, R. Yu, D. Huang, *Compos. Part Appl. Sci. Manuf.* **2016**, *89*, 40–46, DOI 10.1016/j.compositesa.2016.02.006.
[57] J. Wu, Z. Ye, W. Liu, Z. Liu, *J. Ceram. Int.* **2017**, *43*, 13146–13153, https://doi.org/10.1016/j.ceramint.2017.07.007.
[58] S. Zhang, Q. Jiao, Y. Zhao, H. Li, Q. Wu, *J Mater Chem A* **2014**, *2(42)*, 18033–18039.
[59] M. Fu, Q. Jiao, Y. Zhao, H. Li, *J Mater Chem A* **2014**, *2(3)*, 735–744.
[60] G.-M. Li, L.-C. Wang, Y. Xu, *Chin. Phys. B* **2014**, *23(8)*, 088105.
[61] L. Xi, Z. Wang, Y. Zuo, X. Shi, *Nanotechnology* **2011**, *22(4)*, 045707.
[62] S.-C. Chiu, H.-C. Yu, Y.-Y. . Li, *J. Phys. Chem. C* **2010**, *114(4)*, 1947–1952.
[63] R.-B. Yang, P. M. Reddy, C.-J. Chang, P.-A. Chen, J.-K. Chen, C.-C. Chang, *Chem. Eng. J.* **2016**, *285*, 497–507.
[64] Z. Wang, L. Wu, J. Zhou, B. Shen, Z. Jiang, *RSC Adv.* **2013**, *3(10)*, 3309.
[65] X.-J. Zhang, G.-S. Wang, W.-Q. Cao, Y.-Z. Wei, J.-F. Liang, L. Guo, M.-S. Cao, *ACS Appl. Mater. Interfaces* **2014**, *6(10)*, 7471–7478.
[66] Y. Liu, K. Zhao, M. G. B. Drew, Y. Liu, *AIP Adv.* **2018**, *8*, 015223, https://doi.org/10.1063/1.4991448.
[67] L. Landau, E. Lifshitz, *In: Perspectives in Theoretical Physics.* **1992**,, 307-311, 51–65. Elsevier.
[68] D. J. Griffiths, *Prentice Hall, Upper Saddle River, NJ* **1999**, 1-599.
[69] C. Chen, L. Pan, S. Jiang, S. Yin, X. Li, J. Zhang, Y. Feng, J. Yang, *J. Eur. Ceram. Soc.* **2018**, *38*, 1639–1646, https://doi.org/10.1016/j.jeurceramsoc.2017.11.052.
[70] E. N. Zare, P. Makvandi, B. Ashtari, F. Rossi, A. Motahari, G. Perale, *J. Med. Chem.* **2020**, *63*, 1–22, DOI 10.1021/acs.jmedchem.9b00803.

Manuscript received: November 24, 2023

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